

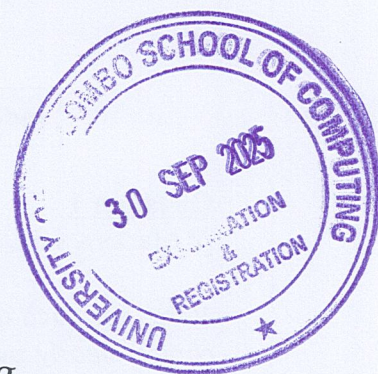


University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

Second Year Examination - Semester I - UCSC AY22 [held in September/October 2025]



IS2207 – Electronics & Physical Computing

(Two (2) Hours)

Answer ALL questions

Number of Pages = 15

Number of Questions = 297

To be completed by the candidate

Index Number:

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Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of this Question paper.
- IV. This paper consists of 2 questions in 15 pages (including the Cover Page).
- V. Answer **ALL** questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Non-Programmable calculators are allowed
- VIII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1 (MCQ)	
2	
Total	

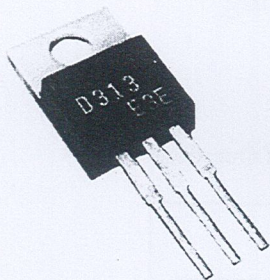
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Question 1

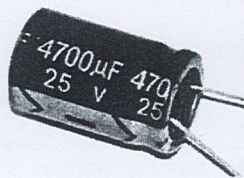
[2 Marks x 10 Questions + 3 Marks x 20 Questions = 80 Marks]

Note: For each question, select the **single best answer** from the given options. Indicate your choice clearly by **circling the option**. If more than one option is selected, no marks will be awarded. For questions that require **justification**, you must provide a brief explanation and relevant calculations. Answers without the required justification will not receive marks. Questions that require justification will carry 3 marks and other questions will carry 2 marks.

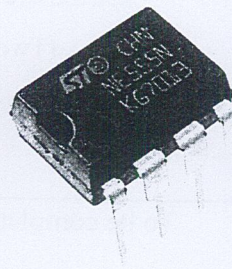
Consider the following electronic components (C1-C4) to answer the questions 1-2.



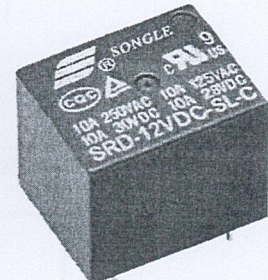
(C1)



(C2)



(C3)



(C4)

- 1) Which option accurately lists the components in the above diagram (may not be in order)?

- (a) Triac, Diode, Transistor, and Optocoupler
- (b) Capacitor, Transistor, Relay, and IC
- (c) Microcontroller, Relay, Triac, and Capacitor
- (d) Diode, Inductor, Resistor, and Switch
- (e) Relay, MOSFET, Diode, and Capacitor

- 2) Which of the following option(s) lists the **active components** in the above diagram?

- (a) C1 & C4
- (b) C2 & C4
- (c) C1 & C3
- (d) C3 & C4
- (e) C1, C3, and C4

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- 3) A steady current of 0.65 A flows through a conductor for 12.0 minutes. What is the total **charge (Q)** (in coulombs, C) that passes through the conductor, and how many **electrons (N)** does this correspond to? ($e = 1.602176634 \times 10^{-19} \text{ C}$)

- (a) $Q = 46.8 \text{ C}$, $N = 2.92 \times 10^{20}$
 (b) $Q = 7.2 \text{ C}$, $N = 4.49 \times 10^{19}$
 (c) $Q = 961 \text{ C}$, $N = 6.0 \times 10^{21}$
 (d) $Q = 120 \text{ C}$, $N = 7.49 \times 10^{20}$
 (e) $Q = 468 \text{ C}$, $N = 2.92 \times 10^{21}$

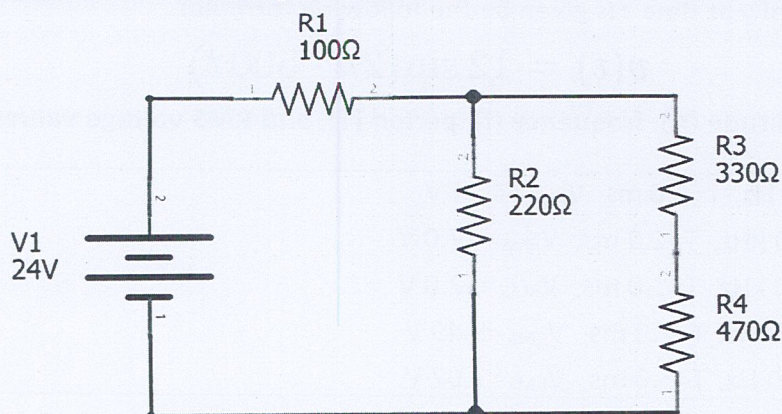
Justify your answer:

- 4) A packet of 3.00×10^{20} electrons passes through a resistor of $R=240 \Omega$ in 15.0 s. What are the values of the total **charge Q**, average **current I**, **voltage** across the resistor **V**, and **power dissipated P**? ($e = 1.602176634 \times 10^{-19} \text{ C}$)

- (a) $Q = 48.1 \text{ C}$, $I = 3.2 \text{ A}$, $V = 769 \text{ V}$, $P = 2.46 \times 10^3 \text{ W}$
 (b) $Q = 4.81 \text{ C}$, $I = 0.32 \text{ A}$, $V = 76.9 \text{ V}$, $P = 24.6 \text{ W}$
 (c) $Q = 7.2 \text{ C}$, $I = 0.48 \text{ A}$, $V = 115 \text{ V}$, $P = 55.2 \text{ W}$
 (d) $Q = 120 \text{ C}$, $I = 8.0 \text{ A}$, $V = 1920 \text{ V}$, $P = 1.54 \times 10^4 \text{ W}$
 (e) $Q = 24.0 \text{ C}$, $I = 1.6 \text{ A}$, $V = 384 \text{ V}$, $P = 615 \text{ W}$

Justify your answer:

- 5) Consider the following circuit.



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Find the **equivalent resistance, total current, voltage across R1, and current through R2.**

- (a) $R_{eq}=220.0\ \Omega$, $I=0.109\text{ A}$, $V_{R1}=9.80\text{ V}$, $I_{R2}=0.055\text{ A}$
- (b) $R_{eq}=320.0\ \Omega$, $I=0.075\text{ A}$, $V_{R1}=7.50\text{ V}$, $I_{R2}=0.065\text{ A}$
- (c) $R_{eq}=180.0\ \Omega$, $I=0.133\text{ A}$, $V_{R1}=13.3\text{ V}$, $I_{R2}=0.06\text{ A}$
- (d) $R_{eq}=272.6\ \Omega$, $I=0.0881\text{ A}$, $V_{R1}=8.81\text{ V}$, $I_{R2}=0.0691\text{ A}$
- (e) $R_{eq}=400.0\ \Omega$, $I=0.06\text{ A}$, $V_{R1}=6.00\text{ V}$, $I_{R2}=0.054\text{ A}$

Justify your answer:

- 6) Three LEDs (Red $V_f=2.0\text{ V}$, Green $V_f=2.2\text{ V}$, and Blue $V_f=3.2\text{ V}$) are connected in series to a 9.0 V supply with a series resistor. Desired current is 15.0 mA . What **resistor value** should be chosen (nearest standard value), what is the actual **current**, and what is the resistor's **color code**?

- (a) $R=110\ \Omega$, $I\approx 14.6\text{ mA}$, Brown–Brown–Brown–Gold
- (b) $R=120\ \Omega$, $I\approx 13.3\text{ mA}$, Brown–Red–Brown–Gold
- (c) $R=150\ \Omega$, $I\approx 10.7\text{ mA}$, Brown–Green–Brown–Gold
- (d) $R=100\ \Omega$, $I\approx 16.0\text{ mA}$, Brown–Black–Black–Gold
- (e) $R=220\ \Omega$, $I\approx 7.27\text{ mA}$, Red–Red–Brown–Gold

Justify your answer:

- 7) A voltage of a signal v at time t is given by the following formula:

$$v(t) = 12 \sin(2\pi \cdot 500 t)$$

What are the **amplitude (A)**, **frequency (f)**, **period (T)**, and **RMS voltage values (V_{RMS})**?

- (a) $A=6\text{ V}$, $f=250\text{ Hz}$, $T=4.0\text{ ms}$, $V_{RMS}=4.24\text{ V}$
- (b) $A=24\text{ V}$, $f=1.0\text{ kHz}$, $T=1.0\text{ ms}$, $V_{RMS}=17.0\text{ V}$
- (c) $A=12\text{ V}$, $f=1.0\text{ kHz}$, $T=1.0\text{ ms}$, $V_{RMS}=12.0\text{ V}$
- (d) $A=12\text{ V}$, $f=500\text{ Hz}$, $T=2.0\text{ ms}$, $V_{RMS}=8.49\text{ V}$
- (e) $A=10\text{ V}$, $f=500\text{ Hz}$, $T=5.0\text{ ms}$, $V_{RMS}=7.07\text{ V}$

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Justify your answer:

- 8) A square wave alternates between 0 V and 10 V. Its period is 1.0 ms, and the pulse width is 0.25 ms. What are the **frequency (f)**, **duty cycle (D)**, **average voltage value (V_{avg})**, and **RMS voltage value (V_{RMS})**?

- (a) $f=500$ Hz, $D=50\%$, $V_{avg}=5.0$ V, $V_{RMS}=7.07$ V
- (b) $f=1.0$ kHz, $D=50\%$, $V_{avg}=5.0$ V, $V_{RMS}=10.0$ V
- (c) $f=1.0$ kHz, $D=25\%$, $V_{avg}=2.5$ V, $V_{RMS}=5.0$ V
- (d) $f=2.0$ kHz, $D=25\%$, $V_{avg}=2.5$ V, $V_{RMS}=2.5$ V
- (e) $f=1.0$ kHz, $D=75\%$, $V_{avg}=7.5$ V, $V_{RMS}=8.66$ V

Justify your answer:

- 9) Three capacitors are connected such that $C_2=20\ \mu\text{F}$ and $C_3=30\ \mu\text{F}$ are in parallel, and this combination is in series with $C_1=10\ \mu\text{F}$. What is the **equivalent capacitance (C_{eq})**?

- (a) $C_{eq}=60.0\ \mu\text{F}$
- (b) $C_{eq}=16.7\ \mu\text{F}$
- (c) $C_{eq}=8.33\ \mu\text{F}$
- (d) $C_{eq}=20.0\ \mu\text{F}$
- (e) $C_{eq}=40.0\ \mu\text{F}$

Justify your answer:

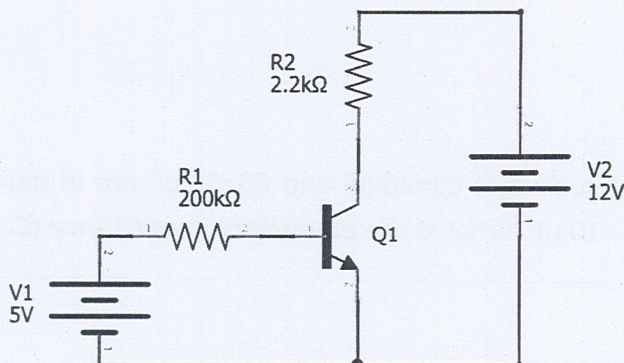
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- 10) A $47\ \mu\text{F}$ capacitor is charged through a $10\ \text{k}\Omega$ resistor from a $12\ \text{V}$ source. What is the **time constant** and the **voltage across the capacitor after $0.2\ \text{s}$** ? Note: the charging equation of a capacitor is $V_c(t) = V(1 - e^{-t/\tau})$.

- (a) $\tau = 0.047\ \text{s}$, $V_c(0.2) = 1.25\ \text{V}$
 (b) $\tau = 0.47\ \text{s}$, $V_c(0.2) = 4.16\ \text{V}$
 (c) $\tau = 4.7\ \text{s}$, $V_c(0.2) = 9.60\ \text{V}$
 (d) $\tau = 0.20\ \text{s}$, $V_c(0.2) = 6.00\ \text{V}$
 (e) $\tau = 0.47\ \text{s}$, $V_c(0.2) = 7.84\ \text{V}$

Justify your answer:

- 11) A BJT transistor has $\beta = 100$, base resistor $R_B = 200\ \text{k}\Omega$, collector resistor $R_C = 2.2\ \text{k}\Omega$, $V_{CC} = 12\ \text{V}$, $V_{BB} = 5\ \text{V}$, and $V_{BE} = 0.7\ \text{V}$ as shown below. What are **base current I_B** , **collector current I_C** , and **collector-emitter voltage V_{CE}** values?



- (a) $I_B = 21.5\ \mu\text{A}$, $I_C = 2.15\ \text{mA}$, $V_{CE} = 7.27\ \text{V}$
 (b) $I_B = 25\ \mu\text{A}$, $I_C = 2.50\ \text{mA}$, $V_{CE} = 6.00\ \text{V}$
 (c) $I_B = 15\ \mu\text{A}$, $I_C = 1.50\ \text{mA}$, $V_{CE} = 8.50\ \text{V}$
 (d) $I_B = 30\ \mu\text{A}$, $I_C = 3.00\ \text{mA}$, $V_{CE} = 5.50\ \text{V}$
 (e) $I_B = 20\ \mu\text{A}$, $I_C = 2.00\ \text{mA}$, $V_{CE} = 7.50\ \text{V}$

Justify your answer:

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- 12) An op-amp is configured as an **inverting amplifier** with $R_{in}=10\text{ k}\Omega$, $R_f=100\text{ k}\Omega$, and input voltage amplitude $V_{in}=0.2\text{ V}$. What are the **gain** (A_v) and the output **voltage amplitude** (V_{out})?

- (a) $A_v=-5$, $V_{out}=-1.0\text{ V}$
- (b) $A_v=-20$, $V_{out}=-4.0\text{ V}$
- (c) $A_v=+10$, $V_{out}=+2.0\text{ V}$
- (d) $A_v=-1$, $V_{out}=-0.2\text{ V}$
- (e) $A_v=-10$, $V_{out}=-2.0\text{ V}$

Justify your answer:

- 13) A 2-input XOR is built using **only NAND gates**. Each NAND has a delay of 10 ns . What is the **overall delay** of the XOR?

- (a) 20 ns
- (b) 30 ns
- (c) 40 ns
- (d) 50 ns
- (e) 60 ns

Justify your answer:

- 14) A TTL output has $V_{OH(min)} = 2.7\text{ V}$, $V_{OL(max)} = 0.5\text{ V}$. A CMOS input requires $V_{IH(min)} = 3.5\text{ V}$, $V_{IL(max)} = 1.5\text{ V}$. Can TTL directly drive CMOS?

- (a) Yes, both HIGH and LOW levels are valid.
- (b) No, because LOW level is invalid.
- (c) No, because HIGH level is invalid.
- (d) Yes, but only if fan-out < 10 .
- (e) Yes, with pull-down resistor.

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- 15) A 4-to-1 MUX has inputs $D_0=0$, $D_1=1$, $D_2=1$, $D_3=0$. Select lines are $S_1=1$, $S_0=0$. What is the output Y ?

- (a) $Y = S_1$
- (b) $Y = D_0$
- (c) $Y = 0$
- (d) $Y = 1$
- (e) $Y = \text{Undefined}$

- 16) A 3-bit synchronous binary counter is clocked at 2 MHz. What are the maximum count value (Count) and the frequency of the MSB output?

- (a) Count = 6, MSB = 500 kHz
- (b) Count = 7, MSB = 250 kHz
- (c) Count = 7, MSB = 500 kHz
- (d) Count = 8, MSB = 250 kHz
- (e) Count = 16, MSB = 125 kHz

Justify your answer:

- 17) A JK flip-flop has $J=1$, $K=1$, and the current state is $Q_n=0$. What is the next state Q_{n+1} after the clock pulse?

- (a) $Q_{n+1} = 0$
- (b) $Q_{n+1} = 1$
- (c) $Q_{n+1} = \text{Undefined}$
- (d) $Q_{n+1} = \text{Same as } Q_n$
- (e) $Q_{n+1} = \text{Toggle only if } J=0$

- 18) A 10 V source is connected in series with a 50 k Ω resistor. A digital multimeter (DMM) with input resistance 10 M Ω is connected across the resistor. What is the measured voltage (V_{measured}) and the approximate measurement error?

- (a) $V_{\text{measured}} = 10 \text{ V}$, Error = 0%
- (b) $V_{\text{measured}} = 9.5 \text{ V}$, Error = 5%
- (c) $V_{\text{measured}} = 9.95 \text{ V}$, Error = 0.5%
- (d) $V_{\text{measured}} = 9 \text{ V}$, Error = 10%
- (e) $V_{\text{measured}} = 8.5 \text{ V}$, Error = 15%

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Justify your answer:

- 19) A 12 V supply is connected across a $120\ \Omega$ resistor. An ammeter with $0.5\ \Omega$ internal resistance is inserted in series. What is the **measured current** (I_{measured}) and **measurement error**?

- (a) $I_{\text{measured}} = 0.095\ \text{A}$, Error = 5%
- (b) $I_{\text{measured}} = 0.0996\ \text{A}$, Error = 0.4%
- (c) $I_{\text{measured}} = 0.1\ \text{A}$, Error = 0%
- (d) $I_{\text{measured}} = 0.105\ \text{A}$, Error = 5%
- (e) $I_{\text{measured}} = 0.09\ \text{A}$, Error = 10%

Justify your answer:

- 20) A microcontroller GPIO pin can source a maximum of 20 mA. You want to drive an LED (forward voltage 2 V) with a 5 V supply. What is the **minimum series resistor**?

- (a) $100\ \Omega$
- (b) $120\ \Omega$
- (c) $150\ \Omega$
- (d) $180\ \Omega$
- (e) $200\ \Omega$

Justify your answer:

- 21) A microcontroller has a 10-bit ADC with $V_{\text{ref}}=5\ \text{V}$. What is the **ADC resolution per step** and the **digital output** for $V_{\text{in}}=3.3\ \text{V}$?

- (a) 4.88 mV/step, D = 512
- (b) 3.2 mV/step, D = 675
- (c) 3.2 mV/step, D = 7
- (d) 5 mV/step, D = 512
- (e) 4.88 mV/step, D = 676

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Justify your answer:

- 22) An Arduino Uno uses `analogWrite(ledPin, 153)` on a pin with 8-bit PWM resolution (0–255). If $V_{cc} = 5\text{ V}$, what is the **duty cycle** and the **average voltage**?

- (a) 66%, 3.3 V
- (b) 60%, 3.3 V
- (c) 15%, 0.75 V
- (d) 60%, 3 V
- (e) 80%, 4 V

Justify your answer:

- 23) The following Arduino code controls an LED:

```
digitalWrite(ledPin, HIGH);  
delay(200);  
digitalWrite(ledPin, LOW);  
delay(800);
```

What is the **blink frequency** and the **duty cycle**?

- (a) 0.5 Hz, 50%
- (b) 1 Hz, 20%
- (c) 1 Hz, 50%
- (d) 2 Hz, 20%
- (e) 2 Hz, 80%

Justify your answer:

- 24) Arduino Uno has a 10-bit ADC, $V_{ref} = 5V$, ADC clock = 125 kHz, 13 cycles per conversion. What are the **resolution**, **conversion time**, and **max sampling rate**?

- (a) 5.0 mV, 80 μs , 12.5 kHz
- (b) 4.88 mV, 104 μs , 9.6 kHz
- (c) 4.0 mV, 200 μs , 5 kHz
- (d) 4.88 mV, 104 μs , 125 kHz
- (e) 10 mV, 130 μs , 7.7 kHz

Justify your answer:

- 25) What is the purpose of the following code?

```
volatile int count = 0;

void setup() {
  attachInterrupt(digitalPinToInterrupt(2), increment, RISING);
}

void loop() {
  Serial.println(count);
  delay(1000);
}

void increment() {
  count++;
}
```

- (a) Counts the number of falling edges on pin 2
- (b) Increment the count by two and print the count every one second
- (c) Prints 0 every second
- (d) Counts the number of rising edges on pin 2 and prints the count
- (e) Generates PWM on pin 2

26) After 10 seconds, what will the value of **ticks** be?

```
volatile unsigned long ticks = 0;

void setup() {
  Timer1.initialize(1000);
  Timer1.attachInterrupt(timerISR);
  Serial.begin(9600);
}

void loop() {
  if (ticks % 1000 == 0) {
    Serial.println("1 second passed");
  }
}

void timerISR() {
  ticks++;
}
```

- (a) 1,000
- (b) 5,000
- (c) 10,000
- (d) 50,000
- (e) 100,000

Justify your answer:

27) Which statement about SPI is **correct**?

- (a) SPI uses a single wire for both clock and data
- (b) SPI slaves are addressed via 7-bit addresses
- (c) SPI cannot be used for multiple slave devices
- (d) SPI does not require a master device
- (e) SPI is synchronous and usually faster than I²C

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28) In physical computing, why might **I²C** be chosen over **SPI**?

- (a) I²C is faster than SPI
- (b) I²C requires fewer pins
- (c) I²C supports higher voltage levels
- (d) I²C does not need addressing
- (e) I²C works only with one master

29) Which platform is best suited for running lightweight **machine learning inference** on-device in physical computing?

- (a) Arduino Uno
- (b) ESP32 (with TinyML)
- (c) Raspberry Pi
- (d) Basic 555 Timer IC
- (e) Passive RC filter

30) Which statement about ADCs is correct?

- (a) Arduino Uno has more ADC channels than ESP32
- (b) ESP32 offers multiple ADC channels with higher resolution than Arduino Uno
- (c) Arduino Uno supports 12-bit ADC, while ESP32 supports only 10-bit
- (d) Both platforms lack analog input capability
- (e) ESP32 requires an external chip for analog inputs

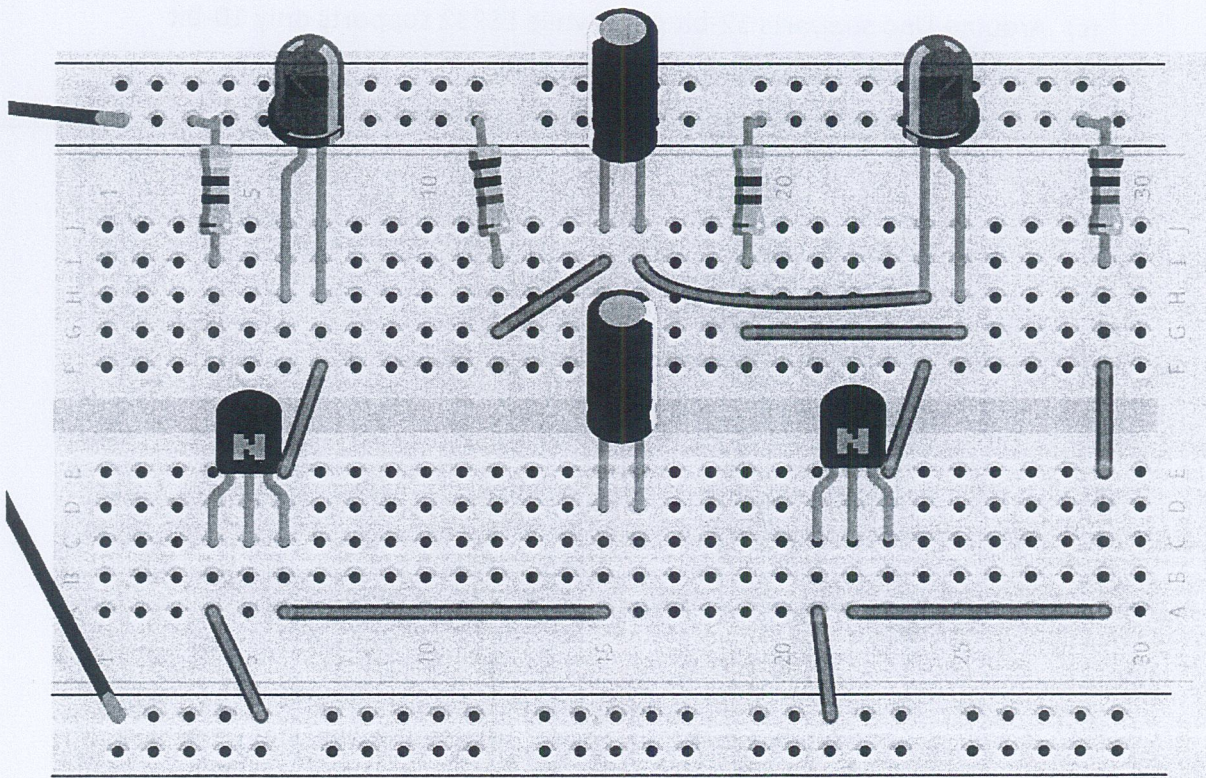
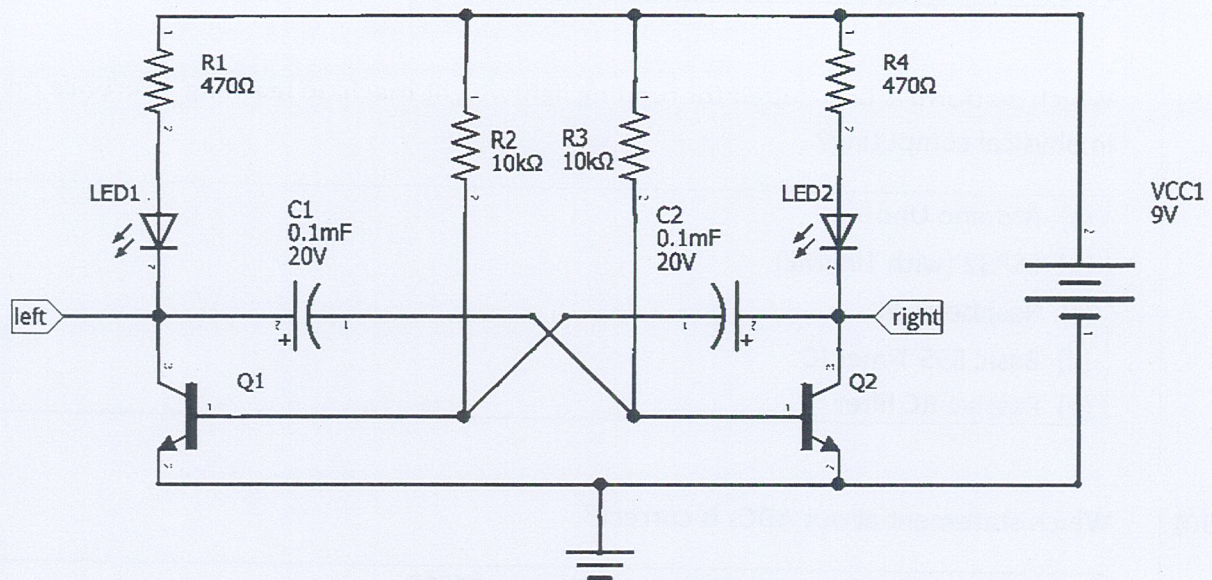
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Question 2

A student has built the following **astable multivibrator circuit** on a breadboard. However, the circuit does not function correctly because some connections are missing. Carefully inspect the given breadboard layout and identify the missing connections (excluding battery connection). Then, complete the circuit by placing the missing jumper wires in the appropriate locations.

[20 Marks]



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Please share your feedback on this question paper in the space provided below.

